CS 6376 / PROJECT PROGRESS

# Drone Swarm Formation Control and Collision Avoidance

Luke Anderson • Dr. Karsai • November 4, 2026

## 1 Work completed

I have a working Python baseline for 20 drones moving from a circle to a grid. A central planner assigns each drone one destination before departure. Each drone then tracks its stored target using its own position and velocity. The planar double-integrator model limits speed and acceleration. Matplotlib plots the flight paths and the closest pair's separation over time.

I ran the same controller with two assignments. With a spacing threshold of 0.750 m, index matching produced 36 pairwise violation episodes. Minimum-total-distance matching produced zero in this transition. Halving the integration step from 0.050 s to 0.025 s preserved both counts. Four automated checks passed, covering controller limits, arrival, assignment, and how violations are counted.

## 2 Challenges encountered

Both runs reached the grid. Arrival alone would have hidden a closest approach of just 0.009 m in the index run. Conversely, the optimized assignment left no violations for an avoidance controller to correct in this example. The comparison therefore needs cases that force an encounter. The present model assumes ideal horizontal acceleration tracking; it does not capture attitude, thrust, or radio behavior. Storing the target locally removes the need for ongoing central commands after delivery.

## 3 Remaining work and timeline

After the update, I will add a heart formation and a two-drone head-on case. In the following two work sessions I will add tracking, avoidance, and hold modes with separate entry and exit thresholds for avoidance. A head-on pair may keep its distance yet stop making progress.

Over the next two sessions I will compare both controllers using identical starts and assignments, recording minimum separation, violation episodes, completion time or timeout, and total travel. For the final demonstration I will use paired results and trajectories to explain where local avoidance helps and where it fails.